

Vibration Suppression Procedure for Equipment WPH249

Based on Mitsubishi Electric MELSERVO-J5 Series

1. Diagnostics Summary & Parameter Mapping

Based on the triaxial vibration diagnostics, the equipment exhibits three distinct vibration mechanisms. We will map these to the appropriate MR-J5 vibration suppression functions:

- **Band A (1.9 - 2.4 Hz): Rigid Body Motion / Machine Cycle**
 - *Characteristic:* Low-frequency global swing, peaks around 2.19 - 2.26 Hz (Upper) and 1.90 - 2.37 Hz (Lower).
 - *MR-J5 Solution:* **Advanced Vibration Suppression Control II (Control 1)** or **Command Notch Filter**.
- **Band B (13.8 - 16.2 Hz): Transmission Harmonics**
 - *Characteristic:* Mid-frequency ripple, likely related to motors/belts/control cycles.
 - *MR-J5 Solution:* **Advanced Vibration Suppression Control II (Control 2)**.
- **Band C (60 - 70 Hz & 145.6 Hz): Structural Modals**
 - *Characteristic:* High-frequency resonance. Upper Z-axis is highly stable at 145.6 Hz. Lower Z-axis shifts to a 60-70 Hz wideband group depending on posture.
 - *MR-J5 Solution:* **Machine Resonance Suppression Filters 1 & 2** and **Adaptive Filter II**.

2. Automatic Adjustment Procedure

Before manually forcing parameters, it is highly recommended to run the servo amplifier's automatic tuning functions. The MR-J5 is equipped to handle complex multi-inertia resonances automatically.

Step 1: Execute One-Touch Tuning

1. Ensure the machine is in a safe position to perform short reciprocating movements.
2. In MR Configurator2, open the **One-touch Tuning** window.
3. Select the **Amplifier command method**.
4. Choose a Response Mode (start with **Basic mode**; if the machine is highly rigid, try **High mode**).
5. Click **Start**. The amplifier will automatically estimate the load inertia and configure the machine resonance suppression filters and advanced vibration suppression control.

Step 2: Enable Adaptive Tuning (For Band C)

If One-Touch Tuning does not fully suppress the 145.6 Hz and 60-70 Hz resonances, enable Adaptive Filter II to track and suppress them dynamically.

1. Set Pr. PB01.0 (Filter tuning mode selection) to **"1" (Automatic setting)**.
2. Set Pr. PB01.3 (Tuning accuracy selection) to **"1" (High accuracy)** for better detection of the shifting 60-70 Hz band.
3. Operate the machine. The amplifier will detect the resonances and automatically populate Pr. PB13 and Pr. PB14.

Step 3: Enable Auto Advanced Vibration Suppression (For Bands A & B)

To automatically target the low-frequency swings (1.9-2.4 Hz and 14-16 Hz):

1. Set Pr. PA24.0 (Vibration suppression mode selection) to **"1" (3 inertia mode)**. *Note: This is required to use two distinct suppression frequencies.*
2. Set Pr. PB02.0 (Vibration suppression control 1 - Tuning mode selection) to **"1" (Automatic setting)**.
3. Set Pr. PB02.1 (Vibration suppression control 2 - Tuning mode selection) to **"1" (Automatic setting)**.
4. Operate the machine normally. The amplifier will estimate the load-side vibration frequencies and populate Pr. PB19 through Pr. PB22 and Pr. PB52 through Pr. PB55.

3. Manual Adjustment Procedure

If the automatic functions fail to fully eliminate the vibrations (especially the posture-sensitive lower Z-axis or the specific transmission harmonics), proceed with manual parameter injection based on your FFT data.

Step 1: Suppress Band C (Structural Modals) manually

We will use standard notch filters to handle the high frequencies.

- **Target 1: 145.6 Hz (Upper Z-Axis, Stable)**
 - Pr. PB13 (Machine resonance suppression filter 1): Set to **145** [Hz].
 - Pr. PB14.1 (Notch depth selection 1): Set to **"0" (-40 dB)** for maximum suppression.
- **Target 2: 60 - 70 Hz (Lower Z-Axis, Wideband Shift)**
 - *Strategy:* Because this peak shifts (63.6Hz to 69.5Hz), we need a wider notch filter centered in the middle of the band.
 - Pr. PB15 (Machine resonance suppression filter 2): Set to **66** [Hz] (Center of the 63-69 Hz band).
 - Pr. PB16.0 (Machine resonance suppression filter 2 selection): Set to **"1" (Enabled)**.
 - Pr. PB16.2 (Notch width selection 2): Set to **"3" (Wide, $\alpha=5$)** to ensure the entire 60-70Hz envelope is covered.

Step 2: Suppress Band B (13.8 - 16.2 Hz Transmission Harmonics)

We will use the second tier of the Advanced Vibration Suppression Control II.

1. Ensure Pr. PA24.0 is set to **"1" (3 inertia mode)**.
2. Ensure Pr. PB02.1 is set to **"2" (Manual setting)**.
3. Pr. PB52 (Vibration suppression control 2 - Vibration frequency): Set to **14.5** [Hz].

4. Pr. PB53 (Vibration suppression control 2 - Resonance frequency): Set to **14.5** [Hz].

Alternative for Band B (Command Notch Filter):

If the vibration persists during constant velocity (not just at settling time), use the Command Notch Filter instead:

- Pr. PB45.0-1 (Command notch filter setting frequency): Set to **"43" (14.7 Hz)** or **"42" (15.6 Hz)**.
- Pr. PB45.2 (Notch depth selection): Set to **"0" (-40.0 dB)**.

Step 3: Suppress Band A (1.9 - 2.4 Hz Rigid Body Motion)

We will use the first tier of the Advanced Vibration Suppression Control II to handle the global machine cycle swing.

1. Ensure Pr. PB02.0 is set to **"2" (Manual setting)**.
2. Pr. PB19 (Vibration suppression control 1 - Vibration frequency): Set to **2.2** [Hz].
3. Pr. PB20 (Vibration suppression control 1 - Resonance frequency): Set to **2.2** [Hz].
4. Pr. PB21 / Pr. PB22 (Damping settings): Leave at initial value **0.00** initially. If the amplitude decreases too quickly and jerks, increase slightly.

Alternative for Band A (Command Notch Filter):

If Advanced Vibration Suppression control is limited by the model control gain (Pr. PB07), you can utilize the Command Notch filter for this ultra-low frequency:

- Pr. PB45.0-1: Set to **"6F" (2.23 Hz)** or **"6D" (2.40 Hz)**.

Step 4: Parameter Validation & Review

After manually setting these parameters:

1. Cycle the power to the servo amplifier to ensure all extension parameters (like Pr. PA24) take effect.
2. Perform a test run mimicking the conditions that generated the initial WPH249 report.
3. Monitor the settling time. If the settling time has increased significantly, verify that your model control gain (Pr. PB07) is still within the permissible bounds formulas outlined for Advanced Vibration Suppression Control.